Research statement

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1. Research during my PhD

Context. My current work focuses on integrable exclusion Markov processes that describe particles evolving on a one-dimensional lattice with hard-core interactions, in particular those that can reach non-equilibrium stationary states. Such systems are particularly rich because they are at the crossroads of four fields of mathematical physics: out-of-equilibrium statistical mechanics (large deviation theory, macroscopic fluctuation theory,...), probability theory (interacting particle systems, integrable probability, KPZ universality class,...), algebraic combinatorics (matrix ansatz method, ...) and quantum integrability (coordinate/algebraic Bethe ansatz, quantum group symmetries,...).

My current objective is to discover new kinds of interesting integrable exclusion Markov processes and study them, in particular their algebraic structure that makes them exactly solvable. The approach that my supervisors and I have adopted toward this goal is to use set-theoretical solutions to the Yang–Baxter equation in order to build processes with novel microscopic dynamics that generalise the celebrated Symmetric Simple Exclusion Process (SSEP).

Twisted symmetric exclusion processes and set-theoretical R-matrices [1]. In this first work, we investigated periodic Markov processes constructed from the simplest types of set-theoretical solutions to the Yang–Baxter equation which are called Lyubashenko solutions. We showed that the associated models are equivalent to a family of twisted SSEP which is a generalisation of the multi-species SSEP where a twist is added on one bond of the ring that can change the species of particles crossing it. We characterised the stationary states and studied the transitions between them when the twist changes.

Integrable multi-species SSEP with reactive particle species [2]. In order to find new integrable processes beyond the twisted SSEP family, we considered simple set-theoretical solutions that are not of Lyubashenko type. We found a new family of integrable processes that generalise the multi-species SSEP in the sense that, in addition to the p SSEP-like particle species called singlets, there are q pairs of species called reactive that are able to evaporate/condense or transform by pairs depending on the interpretation. This family of processes is called (p, q)-SSEP and we proved that they are integrable in the periodic case and in the open case for a general type of boundary reservoir that drives the process out-of-equilibrium. Those reservoirs were obtained by Baxterisation of a solution to the constant reflection equation. In one specific case, we related the stationary state of the (p, q)-SSEP to the one of the usual multi-species SSEP and deduced several mean physical quantities.

Phase transition in an open $(p, 1)$-SSEP [3]. In this work in preparation, we show that the $(p, 1)$-SSEP constructed with two different types of boundary reservoirs introduced in the previous paper displays a boundary driven phase transition. We obtain the non-equilibrium stationary state in a matrix product form and deduce several mean physical quantities.

2. Research project

Beyond local processes: integrable avalanche processes

Integrable exclusion Markov processes that are local, i.e. a transition only changes two neighbouring sites, are well studied. Beyond this setting, an interesting integrable example is the asymmetric avalanche process of Priezzhev, Ivashkevich, Povolotsky and Hu [4], in which there is no restriction on the number of particles per site and where stacks of particles can topple and create an avalanche that may sweep many sites instantaneously. I have obtained a new integrable avalanche process which is built on the totally asymmetric limit of the multiparticle asymmetric diffusion model of Sasamoto and Wadati [5]. In this process, each particle of a site jumps at rate 1 to the next site on its right, carrying with it all the particles stacked above it. In the new avalanche process, such local jumps occur with the addition that cascades of blocks of particles can be triggered across several sites. These multi-site transitions are built from the higher conserved charges of the local process. This avalanche process remains integrable for a one-parameter deformation of the block-jump rates.

My first goal is to study this process and compare it with already known avalanche processes in the literature. Beyond this particular model, I would like to investigate the algebraic structure of non-local processes as well as those without exclusion constraints [6]. Finding boundaries that preserve integrability in such processes is also desirable and could lead to the construction of new tools.

Structure of Zamolodchikov–Faddeev algebras (with E. Feingesicht)

The stationary state of local exclusion processes in one dimension can always be written in a matrix product form, justifying the use of the matrix ansatz method in these systems. The matrices entering the expression of the stationary state satisfy algebraic relations given by the so-called telescopic relations. In the integrable case where the bulk microscopic dynamics of the process is obtained from an R matrix, these telescopic relations originate from the Zamolodchikov–Faddeev (ZF) algebra [7] whose algebraic properties have not been extensively studied. With my supervisors and with Edouard Feingesicht who is a specialist in algebraic structures related to set-theoretical solutions to the Yang–Baxter equation (YBE), we aim at studying the structure of the ZF algebras associated with the Baxterisation of involutive set-theoretical solutions. In particular, I believe that under conditions that remain to be specified, these ZF algebras always admit a PBW-type basis. This would shed light on the algebra associated with the bulk telescopic relations of integrable Markov processes constructed from set-theoretical solutions to the YBE. As for the boundary telescopic relations, this requires in addition the study of the Ghoshal–Zamolodchikov relations which could be the next step in this project.

Earlier-stage projects

Two-dimensional matrix ansatz. No exact stationary state is known for the open TASEP in two dimensions. I want to generalise the matrix ansatz to two-dimensional exclusion processes on various lattice geometries and topologies. By the Eckmann–Hilton argument, composing generators horizontally and vertically is not enough and a genuinely two-dimensional algebraic structure is needed. A natural integrability condition would be given by the tetrahedron equation.

Finite-time probability distribution. The master equation of the ASEP is solved exactly on $\mathbb{Z}$ and on the half-line [8], but not with two open boundaries. I want to determine the finite-time distribution of the open ASEP for simple boundaries and then for arbitrary ones using Bethe ansatz methods. It would be desirable to obtain the solution in a form (contour integrals, determinants,...) allowing the extraction of information about how the system relaxes to its stationary state, in particular in the KPZ regime.

Baxterisation and algebras. I want to find new Baxterisation formulas for solutions to the constant YBE generalising the known ones (Temperley–Lieb, Hecke, Birman–Murakami–Wenzl and n-CB algebras [9]), in particular using Kuperberg's spider algebras [10]. This could lead to physical processes with new types of bulk dynamics. A similar endeavour could be pursued for constant solutions to the reflection equation. Finally, braid representations give knot invariants through Markov traces. I want to explore the possibility of using knot invariants and their skein relations to find new Baxterisation formulas.

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